

CONSTANTA, Romania

WMO: 154800

Lat: 44.2139N

Lon: 28.6456E

Elev: 46

StdP: 14.67

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	15.5	19.8	10.5	9.4	16.8	14.7	11.6	22.1	31.9	25.5	27.7	29.3	10.6	0	0.408

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.1	86.7	73.9	84.5	73.0	82.6	72.1	77.9	82.9	75.7	81.6	74.0	80.2	5.0	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
76.4	137.9	81.2	73.8	126.1	79.7	71.9	118.1	78.2	41.4	83.0	39.2	81.7	37.6	80.4	86.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 23.2	18.7	15.0	DB	10.4	91.3	4.3	3.3	7.3	93.7	4.8	95.6	2.4	97.5	-0.7	99.9	(4)
(5)			WB	9.8	79.2	4.3	2.8	6.7	81.2	4.2	82.8	1.7	84.4	-1.4	86.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	55.2	34.9	37.8	44.2	52.2	62.2	70.5	74.9	74.9	67.0	56.9	47.8	38.2	(6)
(7)		DBStd	15.56	8.69	8.61	6.91	5.83	5.36	4.74	3.89	4.08	5.30	6.57	7.98	8.31	(7)
(8)		HDD50	1590	471	348	205	40	1	0	0	0	0	19	132	374	(8)
(9)		HDD65	4503	934	762	645	386	119	10	0	1	40	259	516	831	(9)
(10)		CDD50	3490	1	6	25	106	379	614	771	770	511	233	67	7	(10)
(11)		CDD65	931	0	0	0	2	32	174	306	306	101	9	1	0	(11)
(12)		CDH74	5771	0	0	2	12	89	889	2221	2235	308	14	1	0	(12)
(13)		CDH80	1134	0	0	0	2	7	138	485	474	27	1	0	0	(13)
(14)	Wind	WSAvg	5.9	6.9	6.8	6.8	5.8	5.4	5.1	5.0	4.9	5.7	6.0	6.2	6.7	(14)
(15)	Precipitation	PrecAvg	17.5	1.4	1.1	1.3	1.2	1.7	1.7	1.4	1.3	1.5	1.5	1.7	1.6	(15)
(16)		PrecMax	27.9	6.1	3.9	3.8	4.3	5.3	5.7	4.9	5.4	5.9	4.3	4.6	7.4	(16)
(17)		PrecMin	8.9	0.2	0.1	0.0	0.1	0.1	0.1	0.0	0.0	0.0	0.1	0.0	0.1	(17)
(18)		PrecStd	4.2	1.1	0.9	0.9	0.8	1.2	1.2	1.1	1.3	1.3	1.1	1.3	1.3	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	56.6	63.4	70.4	74.5	80.4	87.0	90.4	89.8	83.2	75.5	70.1	60.7	(19)
(20)			MCWB	49.2	53.5	54.8	59.6	64.8	71.6	71.8	76.3	67.8	65.0	60.7	53.8	(20)
(21)		2%	DB	52.1	56.4	63.0	68.6	76.3	83.6	86.7	86.8	79.6	71.9	64.4	56.4	(21)
(22)			MCWB	47.0	49.5	52.1	57.4	64.6	71.3	74.0	74.8	68.8	64.2	58.6	52.1	(22)
(23)		5%	DB	48.7	52.0	57.9	64.3	73.5	81.1	84.6	84.5	77.2	68.8	61.1	52.8	(23)
(24)			MCWB	45.3	46.7	49.5	55.6	64.2	70.1	73.3	73.6	67.6	62.9	57.4	49.6	(24)
(25)		10%	DB	45.7	48.2	53.7	60.8	71.0	79.0	82.6	82.5	75.4	66.3	58.4	49.7	(25)
(26)			MCWB	43.4	44.5	47.6	54.1	63.0	69.2	72.5	72.6	67.2	60.8	54.9	47.1	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.6	54.1	56.7	61.4	69.2	76.5	81.2	80.6	73.7	68.5	62.0	55.0	(27)
(28)			MCD B	54.6	61.6	67.6	70.1	75.1	82.1	84.9	85.3	77.3	71.8	67.3	58.7	(28)
(29)		2%	WB	47.6	50.0	52.8	58.8	67.1	73.8	78.3	78.1	71.9	66.2	59.8	52.7	(29)
(30)			MCD B	51.1	55.3	61.1	66.4	73.0	80.4	82.7	83.4	76.0	69.6	63.5	55.7	(30)
(31)		5%	WB	45.5	47.2	50.2	56.6	65.5	72.2	75.9	76.0	70.2	63.9	57.7	50.0	(31)
(32)			MCD B	48.1	51.2	56.6	62.3	71.3	78.6	81.5	82.0	74.7	67.8	60.7	52.4	(32)
(33)		10%	WB	43.7	44.8	48.1	54.9	64.0	70.8	74.1	74.3	68.7	61.7	55.5	47.3	(33)
(34)			MCD B	45.7	47.8	52.8	59.8	69.8	77.0	80.4	80.5	73.7	65.5	57.7	49.7	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	9.8	10.4	11.3	11.2	12.1	12.8	13.1	13.0	12.3	11.2	10.2	10.1	(35)
(36)			MCDBR	13.9	17.0	19.0	17.6	16.1	15.4	15.3	14.6	14.1	14.3	13.7	13.9	(36)
(37)			MCWBR	9.9	10.9	10.7	9.1	8.6	8.7	8.7	9.2	8.1	8.9	8.6	10.0	(37)
(38)		5% WB	MCD BR	12.8	15.7	16.9	14.7	14.0	13.2	12.8	13.1	12.2	12.5	12.8	(38)	
(39)			MCWBR	9.7	10.7	10.2	8.4	8.4	8.5	8.9	9.3	8.4	8.8	8.5	10.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.353	0.382	0.412	0.447	0.424	0.425	0.437	0.447	0.424	0.395	0.372	0.353	(40)	
(41)		taud	2.316	2.244	2.191	2.116	2.271	2.302	2.278	2.242	2.277	2.324	2.328	2.329	(41)	
(42)		Ebn at Noon	237	250	259	259	270	269	264	256	252	243	227	223	(42)	
(43)		Edh at Noon	27	34	41	47	42	41	41	41	37	31	26	24	(43)	
(44)	All-Sky Solar Radiation	RadAvg	409	658	1060	1531	1940	2114	2107	1882	1358	847	479	361	(44)	
(45)		RadStd	61	87	109	148	116	119	107	99	103	79	62	54	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	+0.72	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+196	+121		
(47)	Regional (26 neighbors)	+0.94	N/A	N/A	N/A	N/A	N/A	N/A	-221	+175	+110		

Nomenclature: See separate page